APURBA DAS

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EDUCATION

Central University of Jharkhand, Ranchi, India 2023 – Present
Integrated B.Tech. & M.Tech. in Computer Science and Engineering **CGPA: 9.56/10**

RESEARCH EXPERIENCE

Summer Research Intern, National Institute of Technology Rourkela Summer 2026
Supervisor: Dr. Prasenjit Dey

- Developed **HIRAG**, a hierarchical evidence-localization framework for retrieval-augmented scientific literature understanding.
- Represented scientific papers using **Hierarchical Scientific Document Graphs (HSDGs)** and designed a **Query-Aware Section Selector (QASS)** to localize query-relevant document regions.
- Combined structure-aware candidate expansion and dense semantic retrieval to support evidence localization for scientific question answering.
- Evaluated retrieval quality, evidence localization, question-answering performance, and retrieval efficiency on scientific QA benchmarks.

Summer Research Intern, Banaras Hindu University Summer 2024
Supervisor: Dr. Sachchida Nand Chaurasia, Assistant Professor, Department of Computer Science

- Developed **SCMEF**, a storm-centric multi-CME event-fusion framework for geomagnetic storm intensity prediction.
- Built a temporal forecasting pipeline linking CME events, solar-wind observations, and Dst measurements using time-lagged features and XGBoost.

PUBLICATIONS & PRESENTATIONS

CME-Swin: CME Morphology Learning with Swin Transformer for jGeomagnetic Storm Occurrence Prediction
Accepted for conference presentation at *ICMLDE-2026*; accepted for further processing for publication in *Procedia Computer Science*, 2026.

SCMEF: A Storm-Centric Multi-CME Event Fusion Framework for Geomagnetic Storm Intensity Prediction
Accepted for presentation at *URSI-RCRS 2026*, Paper ID: 1163.
Session: G05 – Machine Learning Methods for Ionospheric Modeling.

HIRAG: Hierarchical Evidence Localization
Hierarchical Evidence Localization for Retrieval-Augmented Scientific Literature Understanding
Manuscript submitted to *Applied Soft Computing Journal*, 2026.

PROJECTS

SUSTAIN-X: Explainable NLP for Climate Adaptation and Urban Sustainability Analysis GitHub

- Developing an NLP-based decision-support system to analyze climate-adaptation policies and urban-sustainability challenges.
- Designed an interpretable Urban Sustainability Stress Index from citizen grievance text, with sentence-level evidence from policy documents and representative complaints.

AI-Based Land-Use Classification of the Delhi-NCR Airshed GitHub

- Built a geospatial deep-learning pipeline using Sentinel-2 imagery and ESA WorldCover labels, constructing a dataset of over 8,000 satellite image patches.
- Trained CNN and ResNet18 models with class-weighted loss to address class imbalance in land-use classification.

Go-Clean: Garbage Detection and Automated Challan System Live Demo

- Developed a YOLOv5-based garbage-detection pipeline using CCTV feeds, with bounding-box visualization, violation logging, and automated challan generation.

X-Ray Image Classification using Deep Learning GitHub

- Developed a multi-class medical image-classification pipeline and compared CNN/LSTM-based architectures using validation metrics.

TECHNICAL SKILLS

Programming: Python, C++, Java, Git
NLP / LLMs: Retrieval-Augmented Generation, Semantic Search, Sentence Embeddings, Dense Retrieval, Document Structure Modeling
Machine Learning: XGBoost, Random Forest, SVM, KNN, Regression, Classification
Deep Learning: PyTorch, TensorFlow, CNN, LSTM, YOLO, Transfer Learning
Libraries / Tools: Scikit-learn, NumPy, Pandas, FAISS, OpenCV, GeoPandas, Rasterio

INDUSTRY EXPERIENCE

AI & ML Intern, Briztech Infotechs Jan 2025 – Mar 2025
Worked on data preprocessing, model training, and model evaluation.

POSITIONS OF RESPONSIBILITY

Vice President, Code Crafter Programming Club 2026 – Present
Co-Technical Secretary, Computer Society of India (CSI) 2025 – Present
Joint Secretary, Code Crafter Programming Club 2025 – 2026
Technical Executive, Code Crafter Programming Club 2024 – 2025

ADDITIONAL ACHIEVEMENTS

- Attendee, National Space Challenge, IIT Kharagpur, 2025.
- Participant, Workshop on Data Analytics, IUCAA, IIT Kharagpur, 2024.
- Participant, GIAN Course on AI in Engineering Business Processes, NIT Rourkela.
- Elite Certificate in Programming in Java, NPTEL, 2024.
- Awarded Token of Excellence in Web Development.